COMP2611 COMPUTER ORGANIZATION
COURSE INFORMATION (SPRING 26)

Teaching Team

- **A team of 8 (1 instructor, 1 instruction assistant and 6 postgraduate teaching assistants)**
- **Our promise: always try our best to support high-quality teaching and learning**

Prof. Alex LAM
Instructor L1&L2

Mr. Colin CHOW Yu Hang
Instructional Assistant

- 4 UGTAs as lab helpers and provide weekly Q&A sessions

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Course Websites and Schedule

- **Course webpage https://course.cse.ust.hk/comp2611/**
  - All course learning materials, visit it regularly for updates
- **https://canvas.ust.hk/courses/67649**
  - Homework/PA submission, grade checking
- **Course Piazza https://piazza.com/ust.hk/spring2026/comp2611**
  - Online Q&A forum, learning community

- **2 x 1.5-hour F2F lectures**
- **1-hour F2F tutorial**
- **1-hour F2F lab**

- **Tutorial and Lab start in week 4 (week of Feb 23 – Feb 27)**
- **Tutorials and labs are useful supplement and nice revision of the lectures**

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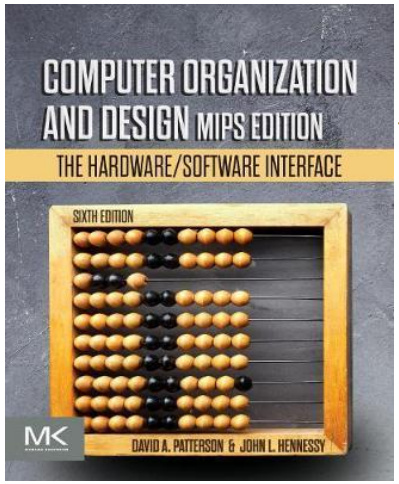
Textbook and Grading Scheme



- **Computer Organization and Design MIPS Edition**
  - 6th edition, ISBN 9780128201091
- **Grading scheme**
  - **4 individual written homework 15% (15%)**
    - Releasing (roughly) in week ~3, week ~5, week ~8, and week ~11, normally due in one or two weeks
  - **Midterm exam 30%**
    - Mar 20 (Friday) 7:00pm, LTA & LTB, week 7
  - **1 individual programming project 15%**
    - Released in week ~9, due in one month (deadline: last day of Spring)
  - **Final Exam 40%**
    - mid/late May

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Course Schedule Overview

	February						
1	1	2	3	4	5	6	7
2	8	9	10	11	12	13	14
3	15	16	17	18	19	20	21
4	22	23	24	25	26	27	28
	March						
5	1	2	3	4	5	6	7
6	8	9	10	11	12	13	14
7	15	16	17	18	19	20	21
8	22	23	24	25	26	27	28
9	29	30	31				
	April						
9				1	2	3	4
9	5	6	7	8	9	10	11
10	12	13	14	15	16	17	18
11	19	20	21	22	23	24	25
12	26	27	28	29	30		
	May						
12						1	2
13	3	4	5	6	7	8	9

- Tutorial and Lab start in week 4
- Week 3: Homework 1 release
- Week 5: Homework 2 release
- Week 7: Midterm (Mar 20 evening)
- Week 8: Homework 3 release
- Week 9: PA release
- Week 11: Homework 4 release

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Academic Integrity

- Course homework and project should be individual work; both ‘provider’ and ‘copier’ will be penalized equally and harshly
- University guideline
- https://registry.hkust.edu.hk/resource-library/academic-integrity

- If you encounter difficulties in learning, do approach us. We can arrange UGTAs to follow up.

Questions left Unanswered

- **In COMP2011 (or similar high-level programming courses), you've learned**
  - Develop a program to solve problem

- **But questions remain as**
  - How does a computer work inside?
  - How can we get electricity to perform abstract tasks like adding and storing numbers?
  - What happens when our programs are executed by a computer?
  - How do we design programming languages and hardware to work together?
  - How do we make computers faster?

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Course Learning Outcomes

- **Upon completion of the course, students are expected to be able to:**
  - Understand the basic concepts of digital logic and build the small circuits involved in computer systems
  - Describe the interaction between software and hardware and instruction set architecture (ISA)
  - Write and execute small programs of a few hundred lines in assembly language
  - Define the basic concepts of modern computer hardware, including processor(datapath, control) and memory
  - Describe the organizational paradigms that determine the capability and performance of computer systems

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Topics to be covered

- **Brief Digital Circuit**
  - Topic 1: Digital Logic (combinational and sequential)
- **Data Representation and Arithmetic**
  - Topic 2: Data Representation (integer, fractional, character)
  - Topic 3: Computer Arithmetic (addition, subtraction, multiplication, division) and ALU
- **Instruction Set Architecture and Assembly**
  - Topic 4: Instruction Set Architecture (ISA) and MIPS Assembly Language
- **Computer Architecture**
  - Topic 5: Processor (datapath and control) and pipeline
  - Topic 6: Memory System (cache, virtual memory)

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